

Machine Learning and Data Mining

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Outline

The problems of Machine Learning (1 week)

Introduction

Estimation

Answering a scientific problem

Course summary

Machine Learning And Data Mining

The nuts and bolts

- ▶ Models
- ▶ Algorithms
- ▶ Practice

Workflow

- ▶ Scientific question and Formalisation
- ▶ Data collection
- ▶ Analysis and model selection

Types of statistics / machine learning problems

- ▶ Classification
- ▶ Regression
- ▶ Density estimation
- ▶ Reinforcement learning

Machine learning

Data Collection

- ▶ Downloading a clean dataset from a **repository**
- ▶ **Scraping** data from the web
- ▶ Conducting a **survey**
- ▶ Performing **experiments**, and obtaining measurements.

Modelling

- ▶ Simple: the bias of a coin
- ▶ Complex: a language model.
- ▶ The model depends on the data and the problem

Algorithms and Decision Making

- ▶ We want to use models to make decisions.
- ▶ Decisions are made every step of the way.
- ▶ Both humans and algorithms can make decisions.

The need to learn from data

Problem definition

- ▶ What problem do we need to solve?
- ▶ How can we formalise it?
- ▶ What properties of the problem can we learn from data?

Data collection

- ▶ **Why** do we need data?
- ▶ **What** data do we need?
- ▶ How **much** data do we want?
- ▶ **How** will we collect the data?

Modelling and decision making

- ▶ How will we **compute** something useful?
- ▶ How can we use the model to make **decisions**?

Problem definition

- ▶ Example: Health, weight and height

Example (Health questions regarding height and weight)

- ▶ What is a normal height and weight?
- ▶ How are they related to health?
- ▶ What variables affect height and weight?

Define a research question

Find a **non-sensitive** variable that we can easily measure via a survey, e.g. related to sleep, smoking, exercise, food, politics, sports, hobbies etc.

- ▶ Discuss in small groups and post suggestions
- ▶ We then vote for what to measure

Data collection

Think about **which variables** we need to collect to answer our **research question**.

Necessary variables

The variables we need to know about

- ▶ Weight
- ▶ Height
- ▶ Dependent: (health/vote/opinion/salary)

Auxiliary variables

Measurable factors related to the variables of interest

Possible confounders

Hidden factors that might affect variables.

Class data and variables

- ▶ The class enters their data into the excel file.



- ▶ Pay attention to the variables we wish to measure

Privacy

- ▶ Is the use of a pseudonym sufficient to hide your identity?

Variables

The class data looks like this

First Name	Gender	Height	Weight	Age	Nationality	Smoking
Lee	M	170	80	20	Chinese	10
Fatemeh	F	150	65	25	Turkey	0
Ali	Male	174	82	19	Turkish	0
Joan	N	5'11	180	21	American	4

- ▶ $\mathbf{X}$: Everybody's data
- ▶ $\mathbf{x}_t$: The t -th person's data
- ▶ $x_{t,k}$: The k -th feature of the t -th person.
- ▶ $\mathbf{x}_k$: Everybody's k -th feature
- ▶ Anything strange with this data?

Raw versus neat data

- ▶ Neat data: $\mathbf{x}_t \in \mathbb{R}^n$
- ▶ Raw data: web pages, handwritten text, graphs, data packets, missing/incorrect values, etc

Types of learning problems

Unsupervised learning (unconditional estimation)

- ▶ Predict the **gender** of an unknown individual.
- ▶ Predict the **height**.
- ▶ Predict the **height and weight**?

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Supervised learning problems (conditional estimation)

- ▶ Classification: Can we predict gender from height/weight?
- ▶ Regression: Can we predict weight from height and gender?
- ▶ In both cases we predict **output** variables from **input** variables

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Variables

- ▶ **Input** variables: aka features, predictors, independent variables
- ▶ **Output** variables: aka response, dependent variables, labels, or targets.
- ▶ The input/output dichotomy only exists in **some prediction problems**.

Course Contents

Models

- ▶ k-Nearest Neighbours.
- ▶ Linear models and perceptrons.
- ▶ Multi-layer perceptrons (aka deep neural networks).
- ▶ Bayesian Networks

Algorithms

- ▶ (Stochastic) Gradient Descent.
- ▶ Bayesian inference.

Reproducibility

- ▶ Modelling assumptions
- ▶ Interactions and feedback

Fairness

- ▶ Implicit biases in training data
- ▶ Fair decision rules and meritocracy

Course material

Books

- ▶ Introduction to Statistical Learning with Python

https://hastie.su.domains/ISLP/ISLP_website.pdf

- ▶ Mathematics for Machine Learning

<https://mml-book.github.io/>

- ▶ Probabilistic machine learning: An introduction

<https://probml.github.io/pml-book/book1.html>

Additional references

- ▶ Probabilistic machine learning: Advanced topics

<https://probml.github.io/pml-book/book2.html>

- ▶ Machine learning in science and society (draft)

<https://github.com/olethrosdc/ml-society-science/blob/master/book.pdf>

Course github

- ▶ <https://github.com/olethrosdc/machine-learning-neuch>

Course structure

Class time

- ▶ Lectures with demos and activities
- ▶ Lab practice
- ▶ Group work

Assessment

- ▶ Class work and assignments (20%)
- ▶ Quizzes (20%)
- ▶ Exam (60%)

Communication

- ▶ ILIAS Forum for **technical** questions
- ▶ Email for **personal** questions
- ▶ Office hours Thursday 13:15-14:00 (by appointment)

Course contents

Theory

- ▶ Estimation
- ▶ Learning and generalisation

Algorithms and Models

- ▶ Bayesian networks and Bayesian inference
- ▶ Neural networks and stochastic gradient descent

Social & Scientific Aspects

- ▶ Reproducibility
- ▶ Fairness
- ▶ Privacy